MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Endocarditis Diagnostic and Management Algorithm

Clinical Features Increasing Suspicion for Infectious Endocarditis (IE)

- Fever (90% pf patients)
- Heart Murmur (85% of patients)
- Systemic symptoms (anorexia, weight loss, night sweats)
- New stroke/neurologic deficits
- New bundle branch block
- New AV block of any degree
- New glomerulonephritis
- Cutaneous features of emboli/immunologic phenomenon
    1. Splinter hemorrhages
    2. Janeway lesions
    3. Osler's nodes)

Consider Early Surgery in Patients with IE

Class I Indications

- Valve dysfunction resulting in symptoms or signs of heart failure.
- IE caused by fungal or highly resistant organisms (eg, vancomycin-resistant Enterococcus, multidrug-resistant Gram-negative bacilli)
- IE complicated by heart block, annular or aortic abscess, or destructive penetrating lesions
- Evidence of persistent infection (manifested by persistent bacteremia or fever lasting > 5-7 days and provided that other sites of infection and fever have been excluded) after the start of appropriate antimicrobial therapy

Class IIa indications

- Recurrent emboli and persistent or enlarging vegetations despite appropriate antibiotic therapy
- Severe valve regurgitation and mobile vegetations > 10 mm

Class IIb indications

- Mobile vegetations >10 mm, particularly when involving the anterior leaflet of the mitral valve and associated with other relative indications for surgery

Risk factors for Infectious Endocarditis

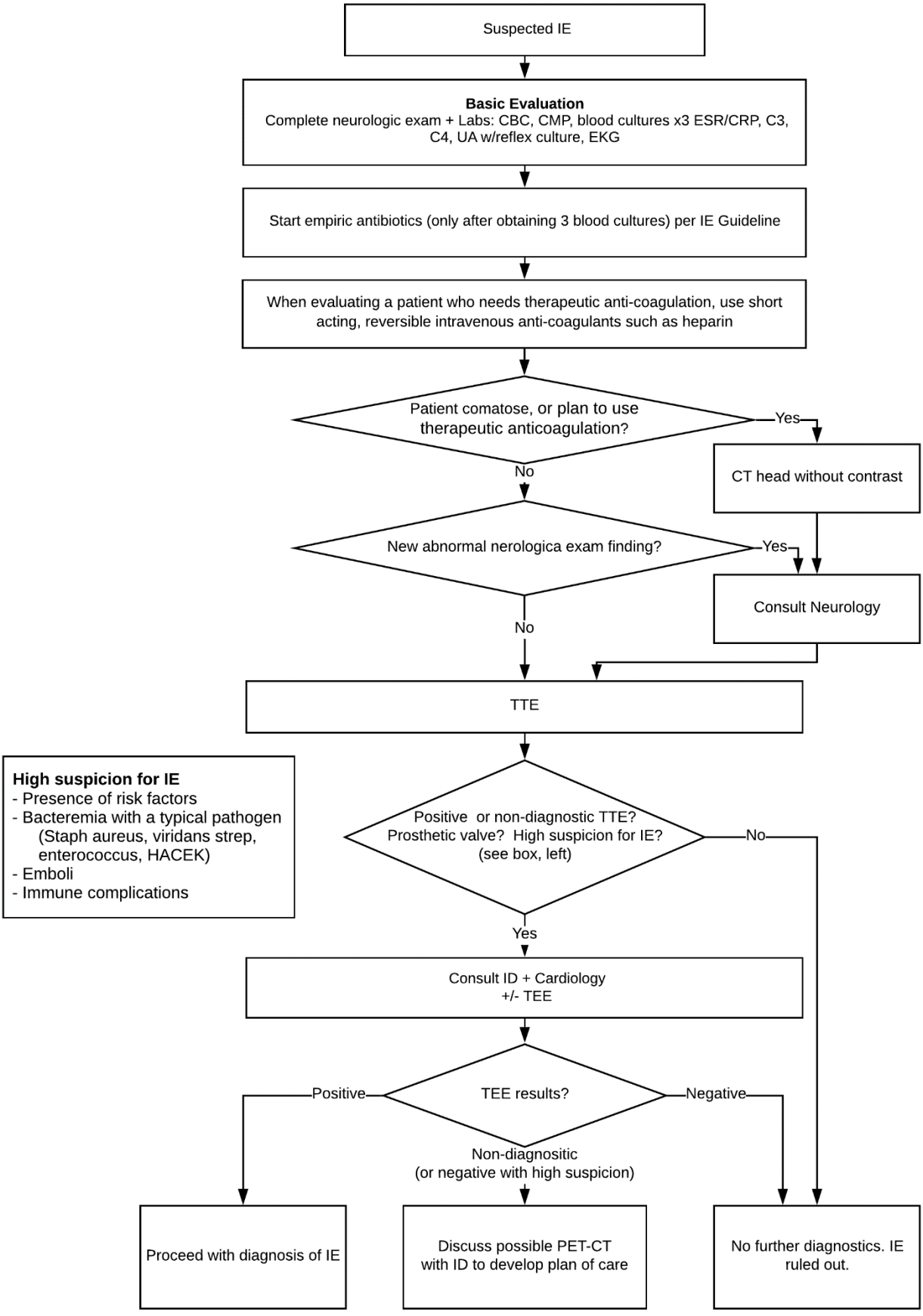
- Prior IE
- Intravenous drug use (IVDU)
- Prosthetic valve
- Implantable cardiac device
- Indwelling central venous access
- Hemodialysis patients
- Poor dentition
- Bicuspid aortic valve
- Transplanted heart with valvulopathy
- Unrepaired cyanotic congenital heart disease or recently repaired with prosthetic material in the last 6 months or repaired cyanotic heart disease with prosthetic material and residual shunt

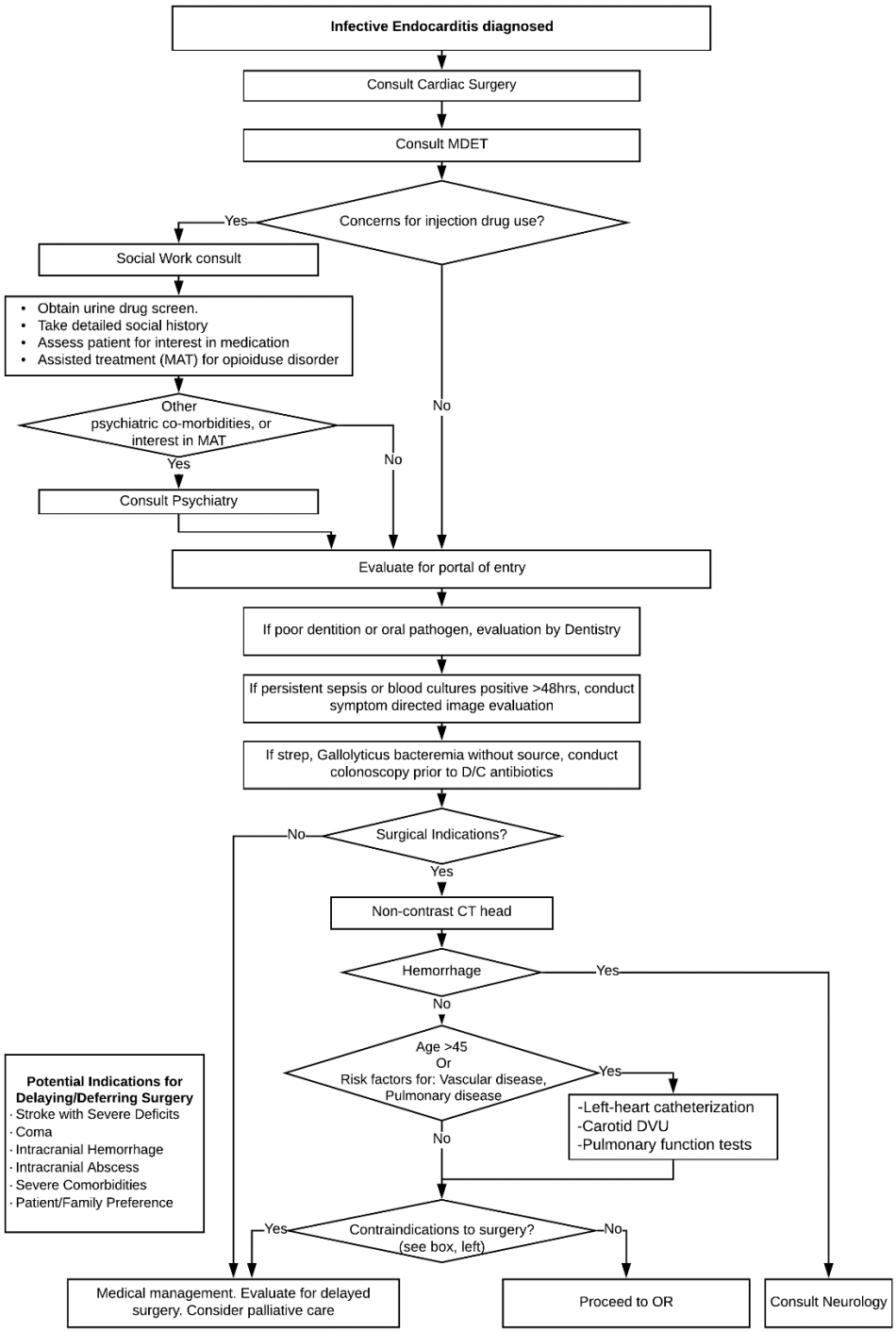
Typical Organisms in Native Valve IE

- Staph Aureus (both MSSA and MRSA)
- Enterococci
- Viridans Streptococci
- Streptococcus Gallolyticus
- HACEK Species

When to Evaluate a Patient for Infectious Endocarditis

Any patient for whom the provider has clinical suspicion for IE
Any patient that has ≥ 2 clinical features
Any patient with a typical organism (see above)


Suspected IE
Basic Evaluation
Complete neurologic exam + Labs: CBC, CMP, blood cultures x3 ESR/CRP, C3, C4, UA w/reflex culture, EKG
Start empiric antibiotics (only after obtaining 3 blood cultures) per IE Guideline
When evaluating a patient who needs therapeutic anti-coagulation, use short acting, reversible intravenous anti-coagulants such as heparin
Patient comatose, or plan to use therapeutic anticoagulation?
Yes
CT head without contrast
No
New abnormal nerologica exam finding?
Yes
Consult Neurology
No
TTE
High suspicion for IE
- Presence of risk factors
- Bacteremia with a typical pathogen (Staph aureus, viridans strep, enterococcus, HACEK)
- Emboli
- Immune complications
Positive or non-diagnostic TTE?
Prosthetic valve? High suspicion for IE?
(see box, left)
No
Yes
Consult ID + Cardiology
+/- TEE
TEE results?
Positive
Negative
Non-diagnositic
(or negative with high suspicion)
Proceed with diagnosis of IE
Discuss possible PET-CT with ID to develop plan of care
No further diagnostics. IE ruled out.



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References		
	Citation	
Revision History		
Date	**Revision**	

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.